INCH-POUND

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9 September 1998
SUPERSEDING
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5 April 1969

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 8766

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: CW Amplifier, 1.0 kW, 4400 - 5000 MHz, 4 integral cavities, individual tuners, waveguide output, coaxial input, permanent-magnet focusing, forced-air cooling.

Outline and dimensions: See figure 1.

Weight: 40 pounds, approximately.

ABSOLUTE RATINGS: 1/

Parameters:	Ef	Eb	Ik	Pi	Load	If (surge)	tk	Iby
Units:	V	kV dc	mA dc	kW	VSWR	a	min.	mA dc
Maximum:	6.8	8.5	600	5.0	2:1	25	----	60
Minimum:	6.0	----	----	----	----	----	3.0	----

Parameters:	Ambient operating pressure	Ambient temperature		Operating temperature 2/		
Units:	atmosphere	°C		°C		
		Oper.	Stg.	Collector	Body	Tuner-Fins
Maximum:	----	+55	+85	+260	+150	+150
Minimum:	0.64	-30	-65	----	----	----

TEST CONDITIONS:

Parameters:	Ef	Eb	tk	Load	Maximum operating temperature
Units:	V	kV dc	min.	VSWR	Collector, body, tuner fins
	6.5	7.5	3.0	1.05 max.	3/ 4/

Cathode leads: See figure 1. RF input connector: Mates with type "N" fitting UG-21D/U or equivalent. RF output flange: Mates with UG-149A/U flange or equivalent. Mounting position: Any. 5/	Cooling: Forced air. 4/ Focusing: Permanent magnet. 6/ Handling caution: 7/

See footnotes at end of table I.

GENERAL:

Preproduction sample approval: Required. 8/

Production Lot: 9/

Dimensions: See figure 1

Marking: 10/ 11/

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Acceptance level	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
Preproduction sample approval tests		12/							
Humidity	---	13/		---	---	---	---	---	---
Tunable frequency	---	14/	F = 4400 to 5000 MHz	---	---	---	---	14	turns
Conformance inspection, part 1		15/							
Visual-mechanical examination	---	---	See figure 1	---	---	---	---	---	---
Heater current	1301	---		0.65	S3	If	7.4	8.2	A
Cathode current	1256	---		0.65	S3	Ik	450	520	mA dc
Body current	1256	---	Power output (2)	0.65	S3	Iby	---	35	mA dc
Power output (2)	4250	16/ 17/	Tuned for high efficiency; F = 4400, 4700, and 5000 MHz; Pd = 50 mW	0.65	S3	Po Po/Pi	1.2 32	--- ---	kW %
Tuner starting torque	4223	---	t = +20°C, approx.	0.65	S3	---	---	30	oz.-in.
Conformance inspection, part 2									
Bandwidth	4271	18/	-3 db (power level at band limits); Power output (2) conditions except F = 4700 MHz	2.5	S1	ΔF	7	---	MHz
Power output (1)	4250	16/ 17/	Tuned for high gain; F = 4400, 4700, and 5000 MHz; Pd = 50 mW	6.5	S1	Po	1.0	---	KW
Tuner stop torque	---	19/	t = +20°C, approx.	6.5	S1	---	---	200	oz.-in.
Conformance inspection, part 3									
Life test	---	---	Group D; Power output (2)	---	---	t	1000	---	hrs
Life test end-points:	---	---	Power output (2)	---	---	Po	1.0	---	kW
Periodic check		20/							
Shock	1042	21/	No voltages	---	---	---	---	---	---
Vibration	---	22/	No voltages	---	---	---	---	---	---
Mechanical tuning fatigue	---	23/		---	---	---	100	---	cycles

See footnotes at top of next page.

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TABLE I. Testing and inspection - Continued.

1/ Refer to MIL-PRF-1, section 3.4.5, "Absolute ratings".

2/ At any ambient temperature, particular care must be taken (see 4/) to maintain the collector, body, and tuner fins below their maximum rated operating temperatures.

3/ The cooling requirements specified or otherwise covered herein (see 4/) shall be effective wherever applicable throughout the testing program herein. The collector temperature shall be measured on the top surface of the collector and the tuner fin temperature shall be measured near the fin shaft. (The thermostat on the collector should be used in an interlock circuit that will remove the beam voltage if the collector temperature becomes abnormally high. This normally closed, Klixon 11041-106-1 or equivalent, thermostat has a switch rating of 50 V dc at 2 amps, or 120 V ac at 1 amp; it opens at 265.6° ± 10°C (510°F ± 18°F), and closes at 237.8°C ± 10°C (460°F ± 18°F)).

4/ Relative to the tube temperature-control requirements specified herein, the following cooling-means accomplishment is recommended. The listed values are approximate, and apply to conditions: sea-level air pressure ≈ 760 mmHg, normal ambient temperature ≈ +20°C.

Air flow		Inlet air pressure	
Collector (ducted)	Body (including tuner fins) (ducted)	Collector	Body
Pound/hour	Pound/hour	Inch-H_2O	Inch-H_2O
815	270	2.0	0.75

5/ The tube may be operated in any position, but the preferred (particularly for fixed-installation operation) mounting position is vertical, cathode end down (see 11/).

6/ Ferromagnetic material shall not be brought within 6 inches of the tube; other magnets, not closer than 12 inches.

7/ Care should be exercised during all handling for installation, storage, and use of the tube, to prevent shock or vibration in excess of the values for which the tube is designed.

8/ The preproduction sample approval requirements in U.S. Army Electronics Command drawing SC-A-46600B hereby replace any qualification requirements referable to the product covered herein. However, the requirements of paragraphs 3.1.2.1, 3.1.2.2, 3.1.3, and 4.4 in drawing SC-A-46600B shall be considered not applicable to the product covered herein.

(Copies of U.S. Army Electronics Command drawing SC-A-46600B, 19 March 1965, "Preproduction Sample Approval In Lieu Of Qualification Requirements In Specifications For Semiconductor Devices And Electron Tubes" may be obtained from the acquiring activity or as directed by the contracting officer.)

9/ A production lot is defined as two (2) months production or seventy (70) tubes, whichever is less.

10/ Tube-designation marking shall be in accordance with MIL-PRF-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only.

11/ Tube identification and tube serial number shall be marked in the most practical and most readily discernible location on the tube, relative to its normal mounted position (see 5/). Also, indication of the sequential alignment of the tuning cavities shall be effected by marking, equivalent to that shown in figure 1.

12/ Although all tests covered herein are required to be performed during preproduction sample approval inspection, the two tests listed under this caption are performed, normally, during preproduction sample approval only. A failure of any one tube in any of the these tests shall be cause for preproduction-sample disapproval.

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TABLE I. Testing and inspection - Continued.

13/ The electron tube shall be subjected to a relative humidity of 90 to 95 percent for 96 hours in accordance with MIL-STD-202, method 103, test condition B. Upon completion of this test, the tube shall meet the requirements for cathode current and power output (2) herein.

14/ The electron tube shall be capable of being tuned smoothly and without discontinuities over the frequency range including the limits specified. The number of turns of the tuner required to tune the tube over the frequency range shall be as specified herein. From lowest to highest frequency, tuning shall be in a clockwise direction.

15/ The acceptance level for the combined defectives for attributes in conformance inspection, part 1, exclusive of "visual-mechanical examination", shall be 1.0 percent. A tube having more than one defect shall be counted as one defective.

16/ With tube synchronously tuned, cavity #3 shall be set approximately one turn high in frequency. Drive power shall be increased to indicated drive power, and then cavity #3 shall be retuned to maximum power output. Minor adjustments of other cavities may be made in order to maximize power output.

17/ The minimum power output condition for power output (1) and (2) shall be available continuously without exceeding the maximum operating temperatures specified under "Absolute ratings" herein, while providing air cooling with an ambient inlet temperature, at sea level, of 45°C - 55°C, and an approximate air flow of 380 pound/hour against a back pressure of 0.45 inches of water, maximum.

18/ A frequency swept driver shall be used to view the power output response during the bandwidth test (see 16/ 17/).

19/ The tuning mechanism of each individual cavity shall withstand the maximum specified stop torque at the extreme ends of the tuning range, without damage.

20/ For periodic tests, sample tubes are be selected at random intervals so that during any twelve month period a minimum of two tubes shall have been successfully tested. In the event of failure of one or more tubes in a sample, these tests are to become a part of conformance inspection, part 2, until two successive lots have successfully completed these tests. After two successive lots have been successfully completed, the periodic tests shall then be eligible to be resumed on a twelve month basis.

21/ The tube shall be subjected (non-operating) to shock test in accordance with MIL-STD-202, test method 213, test condition A except:

Peak value (g's) = 20
Velocity change (VI) = 140.01

After this test, the tube(s) shall meet the power output (2) and cathode current requirements herein.

22/ The tube shall be mounted rigidly on the vibration table, and subjected to vibration, as follows:

a. The tube shall be vibrated under simple harmonic motion through 10 complete scan cycles in each of 3 mutually perpendicular axes, one of which shall be parallel to the beam axis. Each scan cycle shall be of ten minutes duration consisting of a continuously varying vibration frequency between the limits of 10 to 500 to 10 Hz. The vibration frequency shall be divided into 3 steps with peak acceleration as shown below:

G = 10, between 10 and 100 Hz, except a limited displacement of 0.06-inch double amplitude.

G = 5, between 100 and 150 Hz.

G = 2, between 150 and 500 Hz.

b. The waveguide and collector may be supported by a suitable fixture or fixtures such that, when the tube is subject to vibration, the ratio of the peak acceleration, as measured on the flange of the waveguide or on the top center of the collector, with respect to the peak acceleration on the base plate, has a minimum value of 1:1.

c. It is also permissible to secure the tuner shafts to prevent rotation during the vibration test.

After this test, the tube(s) shall meet the power output (2) requirement.

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TABLE I. Testing and inspection - Continued.

23/ The tube shall be operable after each tuning mechanism is subjected to the tuning cycles specified, from one end of the tuning range to the other and back again.

24/ Each electron tube shall be furnished with dust covers for protection of input and output connector elements.

25/ Unless otherwise specified in contract or order, each tube shall, as a minimum protection against moisture and moisture vapor, be sealed within a moisture vaporproof-barrier bag, or equivalent, normally acceptable under packaging requirements of MIL-E-75.

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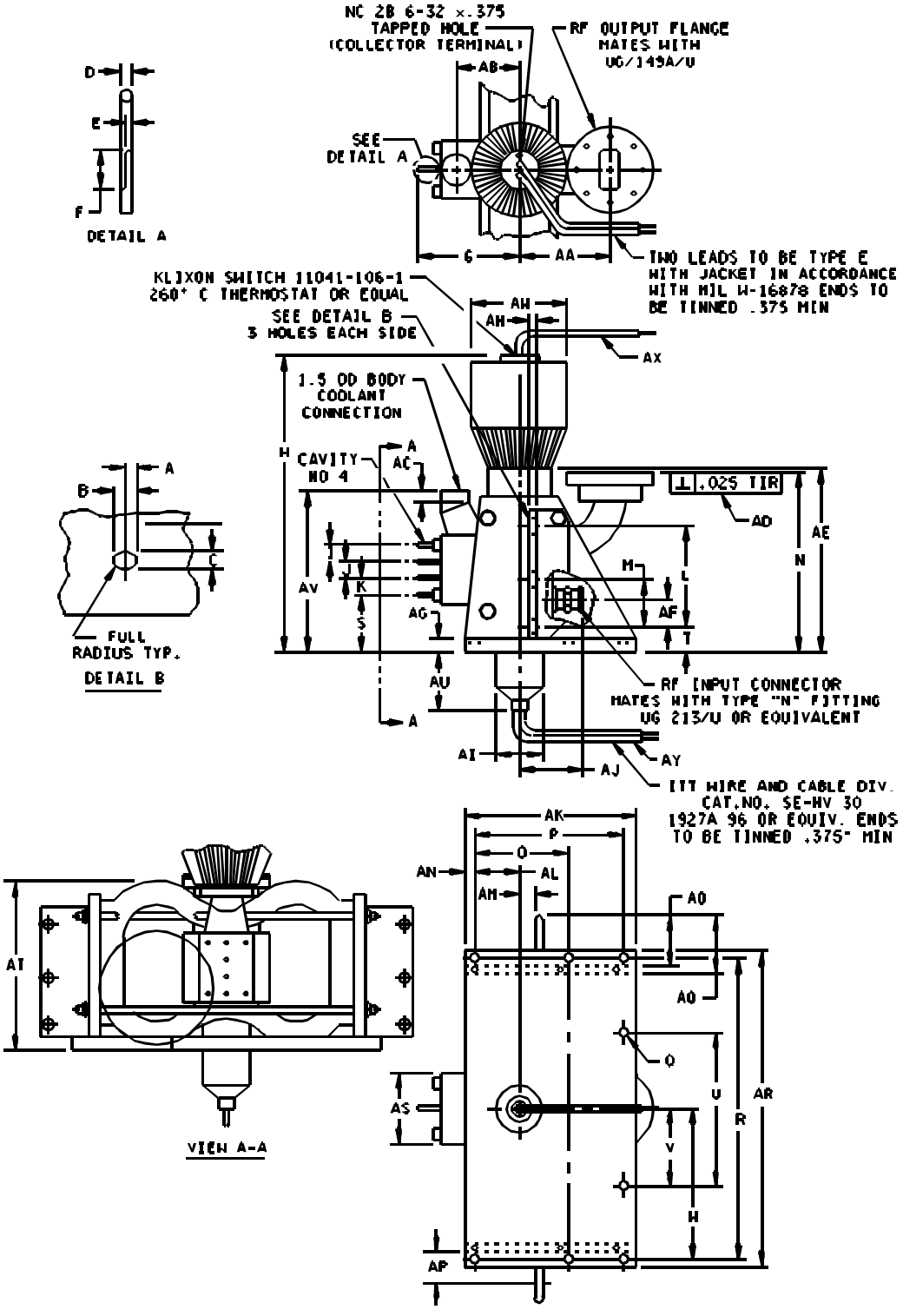


FIGURE 1. Outline and dimensions.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2 (6.5 acceptance level, inspection level S1)				
A	.141	.171	3.58	4.34
B	.297	.327	7.54	8.30
C	.235	.265	5.96	6.73
D	.247	.249	6.27	6.32
E	.170	.180	4.31	4.57
F	.740	.760	18.79	19.30
G	4.481	4.605	113.81	116.96
H	12.875	13.125	327.02	333.37
I	.669	.731	16.99	18.56
J	.669	.731	16.99	18.56
K	.669	.731	16.99	18.56
L	4.735	4.763	120.26	120.98
M	2.360	2.390	59.94	60.70
N	7.375	7.625	187.32	193.67
O	3.990	4.010	101.34	101.85
P	6.865	6.885	174.37	174.87
Q	.297	.327	7.54	8.30
R	12.090	13.010	307.08	330.45
S	2.375	2.625	60.32	66.67
T	.938	1.062	23.82	26.97
U	6.490	6.510	164.84	165.35
V	3.235	3.265	82.16	82.93
W	6.485	6.515	164.71	165.48

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 3(see 20/ of table I)				
AA	3.968	4.032	100.78	102.41
AB	2.875	3.125	73.02	79.37
AC	.485	.515	12.31	13.08
AD	----	.025	----	0.635
AE	7.250	7.750	184.15	196.85
AF	2.375	2.625	60.32	66.67
AG	.485	.515	12.31	13.08
AH	.172	.202	4.36	5.13
AI	2.315	2.375	58.80	60.32
AJ	2.900	3.020	73.66	76.70
AK	7.860	7.890	199.64	200.40
AL	1.985	2.015	50.41	51.18
AM	.485	.515	12.31	13.08
AN	.485	.515	12.31	13.08
AO	1.985	2.015	50.41	51.18
AP	1.422	1.452	36.11	36.88
AQ	.875	.999	22.22	25.37
AR	13.735	13.765	348.86	349.63
AS	2.540	2.600	64.51	66.04
AT	7.375	7.625	187.32	193.67
AU	2.375	2.625	60.32	66.67
AV	6.875	7.000	174.62	177.80
AW	4.063	4.187	103.20	106.34
AX	18.000	----	457.20	----
AY	18.000	----	457.20	----

NOTE: Dimensions are in inches with metric equivalents (mm).

FIGURE 1. Outline and dimensions - Continued.

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Army - CR

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